CAO Short Notes

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1. Define Computer Architecture and Computer Organization

Answer:

Computer Architecture = system ka logical design (instruction set, memory model, CPU behavior). Ye batata hai computer kya karta hai.

Computer Organization = internal implementation (hardware wiring, control signals, components ka arrangement). Ye batata hai computer kaise kaam karta hai.

2. What is the Von Neumann Architecture?

Answer:

Von Neumann Architecture ek computer design hai jisme data aur instructions same memory me store hote hain. CPU sequentially fetch–decode–execute cycle follow karta hai. Isme single memory + single bus hota hai, isi wajah se Von Neumann bottleneck problem aati hai.

3. List the functional units of a computer system

Answer:

Computer ke main functional units:

1. Input Unit (data lena)
2. Output Unit (result dikhana)
3. Memory Unit (data store)
4. Control Unit (operations control)
5. ALU – Arithmetic Logic Unit (calculations aur logic work)

4. What do the terms MIPS and FLOPS stand for?

Answer:

MIPS = Million Instructions Per Second (CPU kitni instructions execute karta hai).

FLOPS = Floating Point Operations Per Second (floating-point calculations ki speed measure karta hai). Dono computer performance measure karne ke metrics hain.

5. Explain the difference between machine language and assembly language

Answer:

Machine language = binary (0s and 1s), directly CPU samajhta hai, human readable nahi hoti.

Assembly language = mnemonics (ADD, MOV), thodi readable hoti hai, lekin assembler se machine code me convert karni padti hai.

6. Describe the working principle of the Von Neumann Architecture with a neat block diagram

Answer:

Iska working Fetch → Decode → Execute cycle pe based hai. CPU memory se instruction fetch karta hai, control unit decode karti hai, ALU execute karta hai.

Block diagram: Input → Memory ↔ CPU (CU + ALU) → Output. Single memory data aur instructions dono store karti hai.

7. Explain the concept of system bus in a computer system

Answer:

System bus ek communication pathway hai jo CPU, memory aur I/O devices ko connect karta hai. Iske 3 types hote hain:

Data bus (data transfer),

Address bus (location batata hai),

Control bus (signals control karta hai). Ye components ko communicate karne me help karta hai.

8. Differentiate between throughput and latency as performance metrics

Answer:

Throughput = ek time me kitna kaam complete hua (e.g., kitne tasks/sec).

Latency = ek task complete hone me kitna time laga.

Simple: Throughput = quantity per time, Latency = delay per task.

9. Convert the decimal number (25) into binary using 2s complement representation

Answer:

+25 ka 2’s complement binary same hota hai normal binary jaisa.

25 = 11001 (binary).

8-bit representation: 00011001.

2’s complement mainly negative numbers represent karne ke liye use hota hai.

10. Convert the hexadecimal number 3A into its binary equivalent

Answer:

Hex ko 4-bit groups me convert karte hain.

3 = 0011

A = 1010

So, 3A (hex) = **00111010 (binary)**.

11. Given a floating-point number, explain how it is represented using the IEEE 754 standard

Answer:

IEEE 754 me floating number 3 parts me hota hai:

Sign bit (positive/negative),

Exponent (scale/power),

Mantissa/Fraction (actual value).

Example: 32-bit me 1 sign + 8 exponent + 23 mantissa bits. Ye format real numbers ko efficiently store karta hai.

12. Compare SISD, SIMD, MISD, and MIMD architectures with suitable examples

Answer:

SISD = Single instruction, single data (normal CPU).

SIMD = Single instruction, multiple data (GPU, vector processing).

MISD = Multiple instruction, single data (rare, fault-tolerant systems).
MIMD = Multiple instruction, multiple data (multi-core processors, distributed systems).

13. Analyze the role of bus interconnection in improving system performance

Answer:

Bus interconnection CPU, memory aur I/O ko fast connect karta hai. Efficient buses parallel data transfer, higher bandwidth aur low delay provide karte hain. Advanced buses (PCIe, system bus hierarchy) bottleneck reduce karke overall system speed aur multitasking performance improve karte hain.

14. Evaluate the impact of floating-point representation errors on computational accuracy

Answer:

Floating-point numbers exact store nahi hote, rounding errors aate hain. Isse precision loss, incorrect comparisons ($0.1+0.2 \neq 0.3$), aur cumulative errors hoti hain. Scientific calculations, finance aur simulations me accuracy affect hoti hai, especially repeated operations me.

15. Design a simple block diagram of a computer system showing all functional units and explain their interaction

Answer:

Block diagram: Input → Memory ↔ CPU (CU + ALU) → Output.

Input data deta hai, memory store karti hai, CPU process karta hai (CU control, ALU calculation), phir result output device par jata hai. Sab units system bus se connected hote hain.

1. Define Central Processing Unit (CPU) and state its main components

Answer:

CPU computer ka brain hota hai jo instructions execute karta hai.

Main components:

1. Control Unit (CU) – control operations
2. ALU – arithmetic & logic operations
3. Registers – fast temporary storage inside CPU

2. What is the function of the Arithmetic Logic Unit (ALU)?

Answer:

ALU CPU ka part hai jo arithmetic operations (add, subtract, multiply, divide) aur logical operations (AND, OR, NOT, compare) perform karta hai. Ye data par actual calculations aur decision-making operations handle karta hai.

3. List the types of instructions used in a CPU

Answer:

CPU instruction types:

1. Data transfer (MOV, LOAD, STORE)
2. Arithmetic (ADD, SUB)
3. Logical (AND, OR, NOT)

4. Control/Branch (JUMP, CALL)
5. I/O instructions (input/output devices handle karne ke liye)

4. What is meant by an interrupt in a computer system?

Answer:

Interrupt ek signal hota hai jo CPU ko current task temporarily stop karke urgent event handle karne ko bolta hai (jaise keyboard input, timer). Task complete hone ke baad CPU wapas previous program continue karta hai.

5. Explain the structure and functions of the CPU with a neat diagram

Answer:

Structure: CPU = CU + ALU + Registers.

Diagram:

Input → Memory ↔ CPU (CU + ALU + Registers) → Output.

Functions: CU instructions control karta hai, ALU calculations karta hai, registers fast data store karte hain. CPU fetch–decode–execute cycle follow karta hai.

6. Describe the instruction cycle and execution process

Answer:

Instruction cycle = Fetch → Decode → Execute → Store.

Fetch: memory se instruction lana.

Decode: CU instruction samajhta hai.

Execute: ALU operation perform karta hai.

Store: result memory ya register me save hota hai. Ye cycle continuously repeat hoti hai.

7. Explain the difference between hardwired control unit and microprogrammed control unit

Answer:

Hardwired CU = fixed logic circuits, fast hota hai but modify karna difficult.

Microprogrammed CU = control memory me microinstructions use karta hai, slower but flexible aur easily update ho sakta hai.

8. Explain the role of program counter (PC) and stack pointer (SP) in program execution

Answer:

PC next instruction ka address store karta hai, execution flow control karta hai.

SP stack ka top address point karta hai, function calls, recursion aur interrupts me push/pop operations manage karta hai.

9. Show how an instruction is executed using different addressing modes with examples

Answer:

Immediate: value direct (MOV A, 5).

Direct: memory address given (LOAD 1000).

Indirect: address pointer se (LOAD (R1)).

Register: register me data (ADD R1, R2).

Indexed: base + offset (LOAD 1000(R1)). Addressing mode operand ka source define karta hai.

10. Illustrate the use of general-purpose and special-purpose registers in instruction execution

Answer:

General-purpose registers (R1, R2) data store aur operations ke liye use hote hain (e.g., ADD R1, R2).

Special-purpose registers jaise PC, SP, IR specific roles handle karte hain—execution control, instruction hold karna, stack manage karna.

11. Demonstrate how the CPU handles an interrupt during program execution

Answer:

Interrupt aate hi CPU current instruction complete karta hai, PC aur state stack me save karta hai. Phir interrupt service routine (ISR) execute karta hai. ISR complete hone ke baad saved state restore karke original program resume hota hai.

12. Analyze the instruction formats and their impact on CPU performance

Answer:

Instruction format me opcode + operand fields hote hain. Short fixed formats fast decoding dete hain (RISC), isse speed badhti hai. Complex formats flexible hote hain but decoding slow hoti hai (CISC). Format size memory use, execution speed aur pipeline efficiency affect karta hai.

13. Compare instruction pipelining and arithmetic pipelining

Answer:

Instruction pipelining: multiple instructions ko stages me overlap karke execute karta hai (fetch, decode, execute).

Arithmetic pipelining: ek complex arithmetic operation (like floating-point multiply) ko stages me divide karta hai.

First overall program speed badhata hai, second specific math operations fast karta hai.

14. Evaluate the effectiveness of pipelining in improving CPU performance, mentioning its limitations

Answer:

Pipelining multiple instructions parallel stages me run karke throughput badhata hai, CPU utilization improve hoti hai.

Limitations: pipeline hazards (data, control, structural), branch delays, dependency issues. Inse stalls aur performance drop ho sakta hai.

15. Design a simple CPU block diagram showing ALU, CU, registers, and data paths, and explain their interaction

Answer:

Diagram: Registers ↔ ALU, aur CU sab control karta hai.

CU instructions decode karke control signals deta hai. Registers data hold karte hain, ALU operations perform karta hai, aur data paths components ke beech data flow manage karte hain.

1. Define memory hierarchy and list its levels

Answer:

Memory hierarchy storage levels ka arrangement hai speed vs cost balance karne ke liye.

Levels:

1. Registers (fastest)
2. Cache memory

3. Main memory (RAM)
4. Secondary storage (SSD/HDD)
5. Tertiary storage (backup devices)

2. What is the function of cache memory?

Answer:

Cache memory CPU aur RAM ke beech fast buffer hoti hai. Ye frequently used data/instructions store karti hai taaki CPU ko RAM access kam karna pade. Isse access time kam hota hai aur overall system speed improve hoti hai.

3. List different cache mapping techniques

Answer:

Main cache mapping techniques:

1. Direct Mapping
2. Fully Associative Mapping
3. Set-Associative Mapping
   Ye decide karte hain ki main memory block cache me kaha store hoga.

4. What is meant by virtual memory?

Answer:

Virtual memory ek technique hai jisme secondary storage (disk) ko RAM ka extension bana diya jata hai. Isse large programs run ho sakte hain even agar RAM kam ho. OS pages swap karke illusion deta hai ki memory zyada available hai.

5. Differentiate between RAM and ROM with respect to characteristics and types

Answer:

RAM = volatile, read/write, temporary storage (types: SRAM, DRAM).

ROM = non-volatile, mostly read-only, permanent storage (types: PROM, EPROM, EEPROM).

RAM fast hoti hai, ROM firmware store karne ke liye use hoti hai.

6. Explain paging as a virtual memory management technique

Answer:

Paging me memory ko fixed-size pages aur frames me divide karte hain. Program ke pages RAM me load hote hain, baaki disk me rehte hain. Page table logical address ko physical address me map karti hai. Isse fragmentation kam hoti hai aur virtual memory efficiently manage hoti hai.

7. Describe the role of MMU in memory management

Answer:

MMU (Memory Management Unit) logical address ko physical address me convert karta hai. Ye paging, segmentation aur protection handle karta hai. MMU ensure karta hai ki process apni memory space me hi access kare, isse security aur efficient memory use hota hai.

8. Explain the concept of cache coherence in multiprocessor systems

Answer:

Multiprocessor systems me har CPU ka apna cache hota hai. Cache coherence ensure karta hai ki sab caches me same data ka consistent version ho. Agar ek CPU data update kare, toh protocols (MESI) baaki caches ko update ya invalidate karte hain.

9. Illustrate direct mapping technique with a suitable example

Answer:

Direct mapping me har memory block ek fixed cache line me hi store hota hai.

Formula: Cache line = Block number % Cache size.

Example: Cache size = 4 lines.

Block 6 → 6 % 4 = 2, toh line 2 me store hoga. Simple but conflicts zyada hote hain.

10. Explain modern memory technologies: DRAM, SRAM, and Flash memory, highlighting their characteristics, advantages, and applications

Answer:

DRAM = slow but cheap, capacitor-based, main RAM me use hoti hai.

SRAM = very fast, expensive, cache memory me use hoti hai.

Flash = non-volatile, data power off me bhi safe, SSD, pendrive, mobile storage me use hoti hai.

11. Demonstrate the use of TLB in address translation

Answer:

TLB (Translation Lookaside Buffer) ek fast cache hai jo recent page table entries store karta hai. CPU pehle TLB check karta hai. Hit hua toh direct physical address milta hai, miss hua toh page table access hota hai. Isse address translation speed fast hoti hai.

12. Compare direct, associative, and set-associative cache mapping techniques

Answer:

Direct mapping: simple, fast, but conflicts zyada.

Associative mapping: block kahi bhi store ho sakta hai, flexible but costly.

Set-associative: dono ka mix, moderate cost aur performance. Most modern CPUs me set-associative use hota hai.

13. Analyze the causes and effects of memory fragmentation

Answer:

Fragmentation memory blocks uneven allocate hone se hoti hai.

Types: Internal (unused space inside block), External (free blocks scattered).

Effects: memory waste, allocation slow, performance degrade. Paging aur compaction techniques fragmentation reduce karte hain.

14. Evaluate different RAID levels in terms of performance, reliability, and cost

Answer:

RAID 0: fastest, no redundancy, low cost.

RAID 1: mirroring, high reliability, costly.
RAID 5: parity-based, balanced speed + safety.
RAID 10: RAID 1+0 combo, very fast + reliable but expensive. Choice use-case pe depend karta hai.

15. Design a memory hierarchy diagram and explain how it optimizes system performance
Answer:
Diagram (top → bottom): Registers → Cache → RAM → SSD/HDD.
Top levels fast but costly, bottom slow but cheap. Frequently used data upar store hota hai, rare data niche. Isse speed aur cost dono optimize hote hain.

1. Define an I/O system
Answer:
I/O system computer ka part hai jo input devices (keyboard, mouse) aur output devices (monitor, printer) ko CPU se connect karta hai. Ye data transfer aur communication manage karta hai between user, devices aur computer system.

2. What is Direct Memory Access (DMA)?
Answer:
DMA ek technique hai jisme I/O devices directly memory se data transfer karte hain bina CPU involvement ke. CPU sirf initial setup karta hai. Isse fast data transfer hota hai aur CPU load kam ho jata hai.

3. List the types of I/O devices
Answer:
I/O devices types:

1. Input devices (keyboard, mouse, scanner)
2. Output devices (monitor, printer, speaker)
3. Storage devices (HDD, SSD, pen drive)
4. Communication devices (modem, network card)

4. What is meant by memory-mapped I/O?
Answer:
Memory-mapped I/O me I/O devices ko memory addresses assign kiye jate hain. CPU same instructions use karke memory aur I/O dono access karta hai. Separate I/O instructions ki zarurat nahi hoti, programming simple ho jati hai.

5. Explain the difference between memory-mapped I/O and isolated I/O
Answer:
Memory-mapped I/O: I/O devices memory addresses share karte hain, same instructions use hote hain.
Isolated I/O: separate I/O address space hoti hai, special IN/OUT instructions use hote hain.
Memory-mapped simple hota hai, isolated I/O memory space save karta hai.

6. Describe interrupt-driven I/O communication technique
Answer:

Interrupt-driven I/O me device ready hone par CPU ko interrupt bhejta hai. CPU current task pause karke data transfer handle karta hai, phir resume karta hai. Isse polling avoid hoti hai aur CPU time efficiently use hota hai.

7. Explain the working of DMA with a neat diagram

Answer:

Diagram: I/O Device ↔ DMA Controller ↔ Memory (CPU sirf control deta hai).

Working: CPU DMA setup karta hai, phir DMA controller directly memory aur device ke beech data transfer karta hai. Transfer complete hone par DMA CPU ko interrupt bhejta hai.

8. Describe the role of I/O interfaces such as USB and PCI

Answer:

I/O interfaces devices ko motherboard se connect karte hain.

USB: universal, plug-and-play, external devices (mouse, pendrive).

PCI/PCIe: high-speed internal interface, GPU, network card, SSD jaise components ke liye. Ye fast data communication enable karte hain.

9. Illustrate the programmed I/O data transfer technique with an example

Answer:

Programmed I/O me CPU directly device status check karta hai (polling). Jab device ready hota hai, CPU data read/write karta hai.

Example: CPU repeatedly keyboard buffer check karta hai jab tak key press detect na ho. Simple but CPU busy rehta hai.

10. Demonstrate how interrupt processing occurs during I/O operations

Answer:

I/O device ready hote hi interrupt bhejta hai. CPU current instruction complete karta hai, state save karta hai, ISR run karta hai data handle karne ke liye. ISR finish hone par CPU saved state restore karke normal program execution resume karta hai.

11. Show how buffering improves I/O performance

Answer:

Buffering me temporary memory use hoti hai data hold karne ke liye. CPU aur I/O device parallel kaam kar sakte hain. Jaise printer buffer me data store hota hai jab tak print ho. Isse speed improve hoti hai aur waiting time kam hota hai.

12. Compare polling, interrupt-driven I/O, and DMA in terms of CPU involvement and performance

Answer:

Polling: CPU constantly check karta hai, high CPU use, slow.

Interrupt-driven: CPU sirf interrupt par react karta hai, better efficiency.

DMA: minimal CPU involvement, fastest transfer. Performance: DMA > Interrupt > Polling.

13. Analyze the need for synchronization techniques like semaphores and spinlocks in I/O systems

Answer:

Multiple processes same I/O resource use kare toh conflict ho sakta hai. Semaphores aur spinlocks synchronization provide karte hain, race conditions avoid karte hain. Ye ensure karte hain ki shared devices safe aur orderly tarike se access ho.

14. Evaluate different I/O performance improvement techniques such as spooling, buffering, and caching

Answer:

Spooling: jobs queue me store (printer), parallel processing possible.

Buffering: temporary memory use, CPU wait kam.

Caching: frequently used data fast memory me store.

Caching fastest hota hai, buffering smooth transfer deta hai, spooling batch processing me useful.

15. Design a block diagram of an I/O subsystem showing CPU, memory, I/O controller, and devices, and explain the data flow

Answer:

Diagram: CPU ↔ Memory ↔ I/O Controller ↔ I/O Devices.

CPU commands deta hai, memory data hold karti hai, I/O controller communication manage karta hai, aur devices data send/receive karte hain. Data flow controller ke through coordinate hota hai.